Interconnection Structures

Condensed Study Notes

Generated: 2026-04-21 07:07 UTC

COMPUTER ARCHITECTURE & ORGANIZATION LECTURE (WEEK 2)

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Interconnection Structures:

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Interconnection structures

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■ I/O module:

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■ Memory:

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Complex Instruction Set Architecture (CISC):

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Instruction set Architecture: CISC, RISC

[Content could not be generated for this section.]

Reduced Instruction Set Architecture (RISC):

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CPU Design Paradigms: RISC vs CISC

RISC and CISC CPU design paradigms differ in instruction length and complexity. RISC uses shorter, simpler instructions with less cycles per instruction, while CISC employs longer, more powerful commands requiring fewer individual instructions. This tradeoff affects code efficiency, execution time, and hardware cost.

Characteristic of RISC –

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CISC vs RISC Design Paradigm Trade-Offs

Both CISC and RISC design methods have inherent trade-offs. CISC aims to minimize instructions per program but increases cycles per instruction, while RISC reduces cycles per instruction at the cost of increased instructions per program.

Characteristic of CISC:

CISC's defining features include single-command instructions (e.g., ADD) that perform complex tasks, reduced number of general-purpose registers, larger instruction sizes, and potentially longer execution times. This design balances complexity with speed, but at the cost of increased program length and memory requirements.

Difference:

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Instruction and Instruction Types

Instructions are defined by their addressing modes (e.g., immediate, register), operand count, and supported operations. Fixed vs variable length instructions impact instruction density and program size. Notable tradeoffs in instruction design include balancing complexity with execution speed, e.g., RISC's simpler instruction set vs CISC's variable-length code.

Word length

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INSTRUCTION SEQUENCING

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References

See Stallings (2010) for an overview of instruction sequencing.

For RISC vs CISC design paradigms, consult Geeksforgeks on computer organization; ScienceDirect's Computer Science topics provide additional context.